Corners challenges

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1 Introduction

The goal of this challenge is to predict the total number of corners that will be scored during a match. To do this, I first built a model using the training dataset. I experimented with multiple distributions and selected a champion model—the one that achieved the highest number of exact predictions. This top-performing model was then applied to the test dataset.

In the test set, I was also given the odds, known as the Asian lines. Using these lines and my predicted number of corners, I estimated the probabilities that the actual number of corners in a match would fall under, exactly on, or over the line. Finally, I determined whether to place a bet on each match, and if so, how many credits to wager. All of my methods and decisions are detailed in the following paper.

2 Training Models

2.1 Methodology

The training dataset included information for every match from 2005 to 2010, with the following fields: `MatchID`, `LeagueId`, `Date`, `HomeTeamId`, `AwayTeamId`, `Home_Goals`, `Away_Goals`, `Home_Corners`, and `Away_Corners`. I created the target variable—the total number of corners—by summing `Home_Corners` and `Away_Corners`.

To train my model, I split the dataset into two parts:

- Matches from 2005 to 2009 were used to train the models.
- Matches from 2010 were used as a validation set to assess model performance.

I treated this problem as a regression task, with the total number of corners as the dependent variable, and the rest of the available data as independent variables. Since the models produced decimal predictions, I rounded the outputs to the nearest integer.

2.1.1 Feature Engineering

To enrich the available data, I created additional features to capture more information about teams and leagues. These included:

- `HomeTeam_AvgHomeGoals`: Average number of goals scored by the home team when playing at home
- `HomeTeam_AvgAwayGoals`: Average number of goals scored by the home team when playing away
- `HomeTeam_AvgHomeCorners`: Average number of corners earned by the home team at home
- `HomeTeam_AvgAwayCorners`: Average number of corners earned by the home team away from home
- `HomeTeam_Last5HomeCorners`: Rolling average of the last 5 home corners earned by the home team

All this features have also been created for the Away team. For the first few matches where historical data was unavailable, missing values in the rolling averages were filled using the next available data.

These features were essential, as the original ones lacked sufficient predictive power. This was confirmed by testing different combinations, which resulted in poor performance when the engineered features were excluded.

2.1.2 Metrics

To evaluate model performance, I used the following metrics:

- **MAE (Mean Absolute Error)**: Chosen because it is less sensitive to outliers than RMSE
- **Number of Exact Predictions**: The count of matches where the predicted number of corners matched the actual number
- **Number of Predictions within ±1 Corner**: To capture near-miss accuracy, which is also informative

2.2 GLM

Generalized Linear Models (GLMs) are used to understand the relationship between explanatory variables and a count outcome—in this case, the total number of corners in a football match. GLMs are flexible in that they allow for different types of distributions for the dependent variable.

2.2.1 Poisson

The Poisson model is often the first choice for modeling count data. It is appropriate when the response variable (number of corners) is a non-negative integer and when the mean and variance of the distribution are assumed to be equal.

However, initial exploration revealed that the assumption of equal mean and variance does not hold well for our data, which led to the exploration of alternatives.

2.2.2 Negative Binomial

To address the overdispersion observed in the data (variance ¿ mean), we used the Negative Binomial model. This model introduces an additional dispersion parameter, α, allowing the variance to be modeled independently of the mean:

This model is especially helpful when there is heterogeneity in the variance across matches or leagues.

2.3 XGBoost

XGBoost is a gradient boosting framework that builds an ensemble of decision trees in a sequential manner. Unlike GLMs, it does not assume a specific parametric form for the relationship between features and the response variable. Instead, it learns patterns directly from the data.In our case XGBoost minimizes the negative log-likelihood of the Poisson distribution. XG-Boost is harder to interpret for the feature importance,

2.4 Model Comparison

Model	**MAE**	**Exact**	**±1**	**±2**	**±3**
Poisson GLM	2.72	11.5%	34.0%	54.3%	70.3%
Negative Binomial GLM	2.72	11.5%	34.0%	54.4%	70.3%
XGBoost (Poisson)	2.72	**11.9%**	33.8%	**54.8%**	70.0%

Table 1: Performance comparison of different models on validation data

From these results, we can observe that there is no significant difference in performance between the methods tested. To address this limited accuracy, we explored new approaches and aimed to improve the XGBoost model.

2.5 XGBoost Improvements

2.5.1 League-Specific Models

The objective here was to build a separate model for each league. This decision was motivated by the observation that football behavior can differ significantly across leagues, and the VMR analysis showed substantial discrepancies between them. A loop was implemented to train an individual model per league.

2.5.2 Feature Selection

To improve the performance of the XGBoost model, I applied feature selection techniques to remove uninformative variables. First, I used Lasso regression, which introduces a penalty that forces some coefficients to zero. The penalty term was tuned through cross-validation. Additionally, I experimented with forward stepwise selection, which incrementally adds features until the AIC stops improving.

2.5.3 Hybrid Model: Negative Binomial + XGBoost

In the hybrid model, I first trained a GLM with a Negative Binomial distribution to predict the dependent variable. Then, I used XGBoost to model the residuals from the GLM. The final prediction was obtained by adding the outputs of both models. This approach leverages the interpretability and distributional robustness of GLMs and the flexibility of XGBoost in capturing non-linear patterns and complex interactions.

2.5.4 Creation of New Features

I engineered new interaction features to enhance the original dataset and better capture underlying match dynamics. These included:

- **Corner-based interactions**: home vs. away average corners, recent form gap, and defensive pressure.
- **Goal-related interactions**: total goals × home corners, and goals vs. away defense.
- **Weighted averages**: combining recent form and long-term averages.
- **Volatility features**: rolling standard deviation of corners to reflect team consistency and unpredictability.
- **Attack vs. defense dynamics**: ratios between offensive and defensive corner performance.

These features were designed to provide more context around team behavior and improve the model's predictive power.

2.5.5 Summary

Model	**MAE**	**Exact**	**±1**
Baseline	2.41	13.21%	38.32%
Baseline + Lasso	2.45	12.9%	38%
Hybrid (NB+XG)	2.21	14.9%	42.2%
Hybrid (NB+XG) + New Features	1.78	18.4%	51%
Hybrid (NB+XG) + Feature selection	1.94	17.1%	47.4%

Table 2: Performance of the different XG-BOOSTs models with one model for each league

The best-performing model is the hybrid approach that combines a NB GLM with XGBoost trained on the residuals, along with the newly engineered features. This result makes sense given that the number of corners is highly variable across matches and leagues — modeling the residuals separately helps capture patterns that a GLM alone would miss.

In this best version, I did not apply any feature selection to the GLM. Although this reduces interpretability, retaining all features appears to improve performance. The newly created interaction features played a crucial role in boosting model accuracy, particularly when modeling residuals with XGBoost.

3 Test on the new data

In order to test our model on the new data, I first had to reconstruct the features used during training. The test dataset only contained the League ID, date, and the IDs of the home and away teams. To address this, I created a function that aggregates team statistics from the training dataset to generate the required features.

Then, I used the best-performing model developed earlier to predict the number of corners. The combined model was applied to the test dataset, with one model used per league, just as in the training phase.

3.1 Estimated Probabilities: Model vs. Bookmaker

In my dataset, I included the predicted number of total corners obtained from my model.

Using these predictions and a Negative Binomial distribution (with the dispersion parameter estimated from the training data), I computed the model-based probabilities for each outcome: P_{over}, P_{under}, and P_{at}. Note that $P_{at} = 0$ when the line is a half-integer, as a match cannot end with a total number of corners exactly equal to such a line.

I did not directly use the bookmaker's implied probabilities in this step, as they are instead incorporated into the final betting strategy via the Kelly criterion. Additionally, bookmakers typically build in a margin — often shading the odds toward the Over side to exploit recreational bettors' biases. This practice can make Under bets appear more favorable than they truly are.

3.2 Betting Decision: Kelly Criterion

To guide my betting decisions, I applied the Kelly criterion, which determines the optimal fraction of the bankroll to wager based on the perceived value of a bet. My approach considers both the model's estimated probabilities and the odds offered by the bookmaker. For risk management, I used a conservative version — the half-Kelly strategy — which stakes only half the amount suggested by the full Kelly formula.

The Kelly criterion is a well-established method in both finance and betting, known for maximizing long-term growth while controlling downside risk. It ensures that higher-stake bets are only made when both the probability and the payout are in strong agreement with the bettor's model.

4 Possible Improvements

There are several ways in which this work could be extended d. Below are a few potential directions:

- **Explore alternative tree-based models:** Techniques such as Random Forests or Bagging.
- **Time series modeling at the team level:** It would be possible to model each team's historical corner count using time series methods such as SARIMA, which can incorporate seasonality and temporal trends.
- **Add more informative features:** Additional contextual variables—such as weather, team formations, match importance.
- **Hyperparameter tuning:** A more thorough hyperparameter optimization (grid search or Bayesian optimization).
- **Model corners per team:** Instead of predicting total corners directly, one could model the expected number of corners for each team separately and then sum the two to get the total.

5 References

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